9. DFT and FFT Algorithms

Objective

This experiment is divided into four simulations:

1.Discrete Fourier Transform

2. Circular Convolution

3. FFT – Decimation-In-Time (DIT) Algorithm

4. FFT – Decimation-In-Frequency (DIF) Algorithm

Experiment 9.1

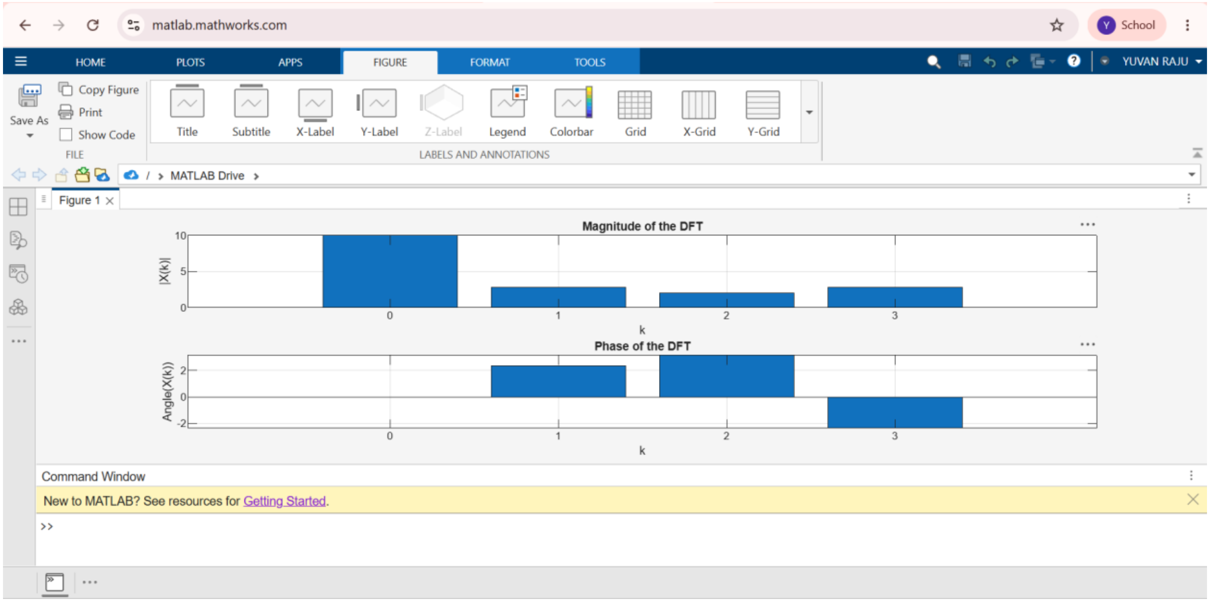
Discrete Fourier Transform

Program

```
% Test Case: Discrete Fourier Transform (DFT) of a Discrete-Time Signal

clc;
clear;
close all;

% Define the discrete-time signal
x_n = [1 2 3 4];

% Compute the Discrete Fourier Transform (DFT)
X_k = fft(x_n);

% Define the frequency axis
N = length(x_n);
k = 0:N-1;

% Compute magnitude of DFT
magnitude = abs(X_k);

% Compute phase of DFT
```

```
phase = angle(X_k);

% Plot magnitude and phase in a single figure
figure;

% Magnitude plot
subplot(2,1,1);
bar(k, magnitude);
xlabel('k');
ylabel('|X(k)|');
title('Magnitude of the DFT');
grid on;

% Phase plot
subplot(2,1,2);
bar(k, phase);
xlabel('k');
ylabel('Angle(X(k))');
title('Phase of the DFT');
grid on;
```

OUTPUT



Experiment 9.2

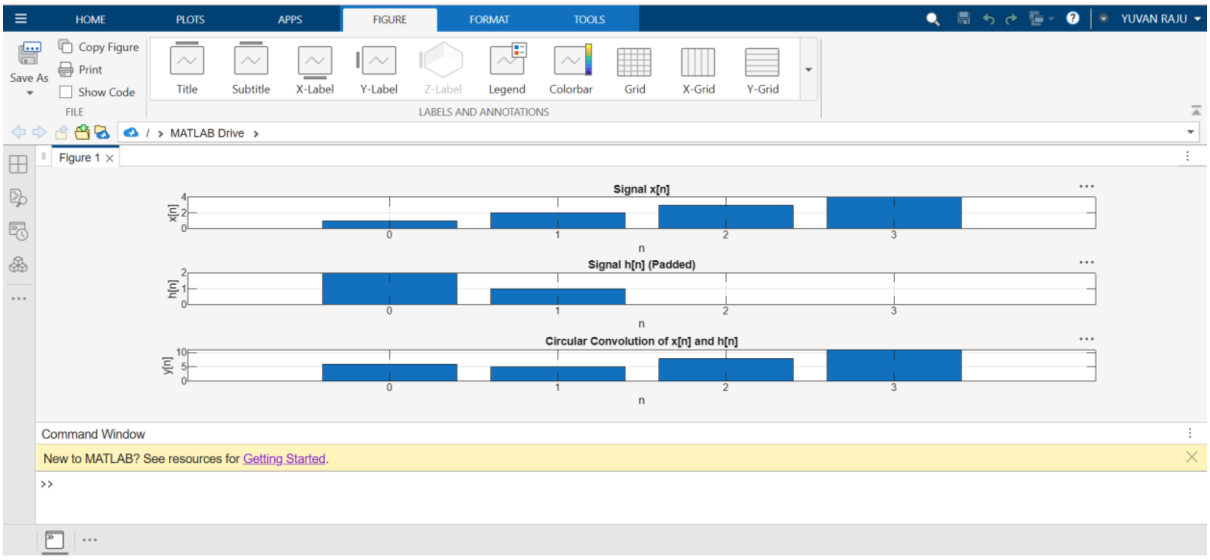
Circular Convolution

Program

```
% Test Case: Circular Convolution of Discrete-Time Signals


clc;

clear;

close all;


% Define the discrete-time signals

x_n = [1 2 3 4];      % Signal x[n]

h_n = [2 1];          % Signal h[n]


% Zero-pad h[n] to match the length of x[n]

N = length(x_n);

h_n_padded = [h_n zeros(1, N - length(h_n))];


% Compute circular convolution

y_n = cconv(x_n, h_n_padded, N);
```

```
% Define time index for output
n_y = 0:N-1;

% Plot all signals in a single figure
figure;

% Plot x[n]
subplot(3,1,1);
bar(0:length(x_n)-1, x_n);
xlabel('n');
ylabel('x[n]');
title('Signal x[n]');
grid on;

% Plot h[n] (padded)
subplot(3,1,2);
bar(0:N-1, h_n_padded);
xlabel('n');
ylabel('h[n]');
title('Signal h[n] (Padded)');
grid on;

% Plot circular convolution
subplot(3,1,3);
bar(n_y, y_n);
xlabel('n');
ylabel('y[n]');
title('Circular Convolution of x[n] and h[n]');
grid on;
```

OUTPUT


HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
YUVAN RAJU
Save As
Copy Figure
Print
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Figure 1
Signal x[n]
x[n]
n
Signal h[n] (Padded)
h[n]
n
Circular Convolution of x[n] and h[n]
y[n]
n
Command Window
New to MATLAB? See resources for Getting Started.
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